Post-Disaster Damage Assessment and Rescue Guidance

CSE499B – Section 13, Group 5, Update 2

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Weekly Update:

The ablation sprint closed without any single change reaching our phase gate (mIoU above 0.52 and Destroyed-class IoU above 0.30 on two seeds): the best recipe, R9 (class-balanced sampler + copy-paste), reached 0.3001 Destroyed IoU on one seed and 0.2941 on its second. An error analysis showed that 59.5% of the destroyed pixels R9 misses are predicted as background, so the model fails to *find* rubble rather than to grade it. We therefore moved to structural changes: (1) feeding the before-disaster image as well (6 input channels, R11), and (2) a find-then-grade pipeline (R16c) in which SAM 3 outlines buildings on the before-image and a Siamese ResNet-18 grades each building from its before and after crops. On the VLM side we tested our fine-tuned Qwen2.5-VL-7B at full image size, re-trained it briefly at that size, fed it building counts as text, and benchmarked an SQL counting agent and an INT8 edge model. A data audit then found that 40 of our 645 validation pairs have their before-image shared with a training pair (xBD's `socal_fire`/`woolsey_fire` twins); every score below is reported both on all 645 pairs and on the 605 clean ones.

Results Summary

Run	Change (8-view TTA where noted)	mIoU	Destroyed IoU	Destroyed IoU, clean 605
R1	Sprint champion (ResNet-50, phased loss, 768 crops), TTA	0.5440	0.2822	0.2689
R9	+ class-balanced sampler + copy-paste, TTA	0.5440	0.3001	0.2874
R11	+ sampler + before-image as input (6 channels), TTA	**0.5875**	0.2965	0.2692
R11s7	R11, second seed, TTA	**0.5913**	0.2928	0.2652
R16c step 2	SAM 3 + R11 pieces, stronger grader (grader seeds 42/43)	0.5869 / 0.5852	0.3159 / 0.3106	0.2980 / 0.2935
R16c step 5	SAM 3 outlines only, grader trained on them (seeds 42/43)	0.5520 / 0.5522	**0.3347** / **0.3242**	**0.3161** / **0.3051**

All on the same 645 validation pairs at full resolution. "Clean" drops the 40 leaked pairs. Only the R16c step 5 map clears 0.30 on clean data, on both grader seeds; a third seed is running.

Abrar Mohammed Tanzim Alam – Structural Runs, VLM Track & Orchestration

1. **Segmentation (phase gate):**
   - Closed the sprint (R1–R10 plus second seeds) and ran the error analysis (E1) that set the order of the structural runs.
   - R11 (before + after input) gave the best mIoU so far (0.5875 with TTA) and cut the share of missed destroyed pixels called background from 60% to 30%. Adding 1,619 extra xBD tiles (R12) changed nothing. A per-event breakdown showed that the 2023 Turkey earthquake holds 37% of the validation set's destroyed pixels and that R11 scores 0.000 there.
   - Built R16c, the find-then-grade pipeline, in five steps: a perfect grader on the found buildings would reach 0.61 Destroyed IoU, so the grader, not the finding, is the bottleneck. A stronger grader passed the gate on all 645 pairs (0.316 and 0.311), and a grader trained on SAM 3's own outlines reached 0.335 and 0.324 with SAM 3's map alone.
2. **Leak check and clean re-scoring (E5):** confirmed the 40 leaked pairs found in Ridita's audit with a pixel-hash check of all 20,988 training and 7,278 benchmark images, and re-scored every affected result. The leak lowered the after-image-only U-Nets by 0.012–0.013 Destroyed IoU and the before + after U-Nets by 0.027. On the benchmark, 186 of 10,675 optical questions (1.7%) use a copied image; removing them moves any score by at most 0.7 points.
3. **VLM track:**
   - Re-scored all six benchmark tasks with greedy decoding: our 7B averages 36.1% on the optical questions, against 31.2% for the base model and 40.4% for the paper's fine-tune. At the paper's full image size it gains 4.5 points on disaster type (71.1 to 75.6) and 7.8 on relational reasoning (23.3 to 31.1).
   - Ran a top-up training of the 7B at full size (28 optimizer steps on two T4s over four sessions); its six-task evaluation runs this week.
   - Built a per-building table from R11's predictions on the benchmark; counting from it scores 32.96% on optical questions (32.47% clean), against the base model's 34.2%.
4. **Edge and reporting:** an INT8 U-Net that is 4× smaller (33 MB) and 1.43× faster on CPU, losing only 0.0003 Destroyed IoU; 1,085 verified teacher answers for distilling the 7B into a 3B model; a survivability heatmap whose circles fall on damaged buildings 73.6% of the time.

Ridita Afrin Riya – Data Audit & Offline Guideline Search

1. **DisasterM3 vs. BRIGHT optical audit:** found that 40 validation pairs (8.6% of the validation set's destroyed pixels) are near-copies of training pairs, which led to the E5 re-scoring. Showed that BRIGHT adds no destroyed examples for us (all its after-images are radar), that its Turkey, Morocco and Libya tiles must never enter training (they are the before-images of 66 validation pairs), and that 52% of before/after pairs are misaligned by more than 2 pixels.
2. **Offline guideline search:** built a local, cited search over 405 sections from six sources (EMS-98, INSARAG Vol. II and III, Sphere, xBD, IWG-SEM), combining MiniLM embeddings and BM25 keywords by reciprocal rank fusion, so the report writers can cite guidelines without internet access.

Aryan Sami – Segmentation-to-VLM Counting Study

1. **TEOChat-style grounding in the prompt:** following TEOChat (ICLR 2025), which found that a specialist U-Net localises buildings far better than a generalist VLM, built a pipeline that turns a segmentation mask into a building count plus normalised bounding boxes (0–1000) and injects them into the VLM's prompt.
2. **Counts in the prompt:** asked our fine-tuned 7B 1,703 optical counting questions with and without these findings. Given the correct count it scores 96.5%; from the image alone, 15.7%. The counting gap is therefore in *seeing* the buildings, not in reasoning about counts, which supports feeding segmenter counts to the VLM as text.

Abdullah Al Noman – SQL Counting Agent

1. **Text-to-SQL agent:** built the agent that answers counting questions by writing SQL over the per-building table; both tested models wrote valid SQL for all 100 sampled questions.
2. **Graded test (H2b):** ran the graded test of four small models (each in 4-bit and fp16) on 297 open questions. No model reached the 90% bar (best 73.9%), so the system will use fixed, pre-written queries.

Tamanna Akter Mou – Building-Crop Inspection Prototype

1. **Agentic vision pipeline:** prototyped a pipeline that crops each building at full resolution and asks the VLM about it, with a PaliGemma 2 baseline. A key lesson: when the VLM is shown the segmenter's class, it agrees almost every time, so agreement cannot be used as accuracy. Scores will be reported after a clean re-run against ground truth.

Next Week's Strategic Roadmap

- **Phase gate:** decide the clean-data result from the third grader seed of R16c step 5, and group all future train/validation splits by before-image so twins never cross the split.
- **VLM:** score the top-up 7B on all six tasks and keep it only if disaster type and relational reasoning rise with no other task dropping more than one point; then distil the better 7B into a 3B model.
- **Report system:** connect the offline guideline search to the DisasTeller report agents running on our local models, and start the offline demo.